Supplementary Material: Reliable Strictly Local Floorplan Editing under Imperfect Controls

Anonymous CVPR submission

1. Scope and Evidence Populations

This supplement expands the protocol and audit boundary. It does not add a new claim. Table 1 lists the populations used by the main paper. Oracle, predicted, identity-fallback, generation, and free-text rows are never pooled.

Table 1. Evidence populations and exposure.

Evidence	Population	Exposure boundary
Oracle anchor	CubiCasa5K test503, 2,479 edits, five operations, five repeated seeds	Target-derived controls; non-deployable
Predicted reasonableness	CubiCasa5K val491, 491 cases per seed, five repeated seeds	Target-free inference; validation
Operation safety	738 typed prompts, 438 supported and 300 abstention	Development-exposed test rows; not held out
Semantic diagnostic	374 supported prompts from an older val191 model	Diagnostic only; not the selected checkpoint
Composition audit	12,395 strict-locality rows	Deterministic policy arms; post hoc budget audit
External context	HouseMind-E, Raster2Seq, CubiCasa5K	Heterogeneous protocols; no leaderboard

2. Outside Contract

The outside domain is

$$O = P \wedge \neg R \wedge \neg D_1(W_0 \vee W_1), \tag{1}$$

where P is the footprint, R the declared edit region, W_0 and W_1 the pre/post wall support, and D_1 one-pixel dilation. The audit compares pixel, room-label, room-mask, polygon, wall, opening, and unaffected-edge content exactly. The changed-control mask is an anchor for the edit and cannot expand O. A violation is a contract failure even if aggregate Delta-IoU is high.

3. Metric Separation

3.1. Edit Support versus Geometry

Delta-IoU and Delta-MSE measure binary change support under aligned metric definitions. They do not establish that the requested room class dominates the changed pixels. The semantic diagnostic therefore asks a different question: does the requested class or family appear, disappear, or dominate the declared change? The diagnostic uses only supported prompts and retains every failure.

3.2. Raw versus Functional Watertightness

Raw fpeval watertightness is 0.816701 on val491. Functional watertightness is 0.997963. Of 491 footprints, 401 are single-component and pass both raw and componentwise checks. The remaining 90 are multi-component footprints outside the standard single-region FP-Eval boundary. The component-aware value is a declared diagnostic, not a replacement for the raw number.

3.3. Safety versus Semantic Success

Supported execution, unsupported abstention, object-delete contract checks, and ghost-state checks are safety or state properties. They are not semantic success. All 438 supported development-exposed prompts execute and all 300 unsupported or ambiguous prompts abstain in the selected audit. The older val191 semantic rates remain 0.92, 0.55, 0.20, 0.5085, and 0.00 for room addition, room type, room removal, opening deletion, and furniture deletion.

4. Statistical Units

Oracle intervals resample floorplans with replacement and average all edits inside a sampled floorplan. Repeated seeds do not create additional independent plans. Paired comparisons use one mean difference per unique floorplan and Holm adjustment over five operations. The seed19 trace audit recomputes every oracle row with zero IoU error. Other seeds are retained as reproducibility checks.

The composition audit compares deterministic routing policies on a fixed row set. Its matched-budget analysis is post hoc. Typed routing improves Delta-IoU by 0.048656

and changes GED by -0.061396 with no added learned parameters, while consuming 3.1925 ms versus 0.2666 ms mean routing time. Efficiency is not claimed.

5. Predicted-Control Behavior

Figure 1 reports safety, structural plausibility, and semantic realization as separate axes. The safety bars are near the ceiling while raw watertightness and semantic success expose different failure modes. This is the intended reading of the figure: a high safety rate is not evidence of a high semantic success rate.

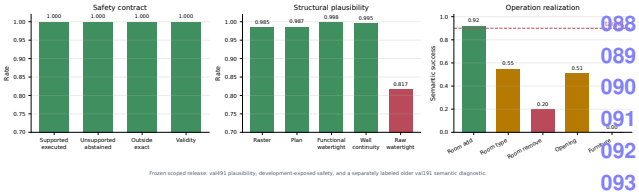


Figure 1. Safety, structural plausibility, and semantic operation realization are separate axes. The semantic panel uses an older val191 model; the safety prompts come from a development-exposed test manifest. Neither is held out.

6. Failure Taxonomy and Case Selection

Figure 2 retains failed supported prompts. The left panel reports success and failure counts with the actual prompt denominator. The image panels show one deterministic failed case from the val191 raster cache. The selection rule prioritizes room deletion, then room type, opening, and furniture, and takes the first lexicographic failed row with an existing raster. The case identifier and failure reason are recorded in `outputs/reports/floorplan_figure_case_manifest_v1.json`.

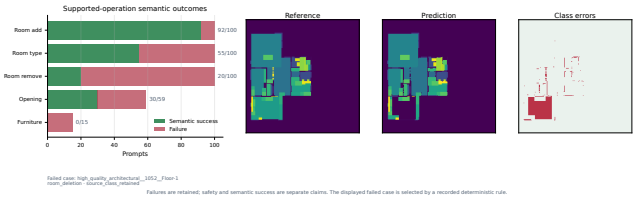


Figure 2. Supported-operation outcomes and a retained failure. Counts use supported-prompt denominators. The displayed case is selected by a recorded rule; it is not selected because it is representative of a favorable outcome.

7. Held-Out Availability

No fresh held-out split is available for a scoped claim. All test cases were used in full predicted-completion inference, so excluding prompt rows would not create an unopened model split. The optional extension is reported as `optional_generalization_not_evaluated`. It does not block this release. A future held-out claim must freeze the checkpoint, configuration, prompts, taxonomy, evaluator, and thresholds before access, execute once, retain negative outcomes, and report the extension separately from the main tables.

8. Artifact and Auditing

The anonymous artifact contains source, relevant tests, selected configurations, manuscript sources, tables, figures, and registry-bound evidence. It excludes datasets, checkpoints, raw arrays, caches, and third-party source trees. The source registry, provenance audit, freeze configuration, freeze audit, artifact manifest, and checksum list are machine-readable. The local submission checklist separates completed package checks from actions requiring author authorization in OpenReview.

LLM Usage Statement

Large language model assistance was used for code editing, artifact auditing, LaTeX formatting, and language editing. Numerical claims were verified against the frozen registry and generated tables. No held-out result, execution receipt, or submission receipt was fabricated. The authors remain responsible for the final manuscript, data use, and external submission.